Universal Analog-Hawking Saturation Ratio $\rho = c/12$ for Unitary Diagonal Minimal-Invariant-Partition CFTs

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Abstract

We prove that the analog-Hawking saturation ratio $\rho := (2\pi)^{-2}\int_0^\infty S_{\mathrm{BW}}(u)\,du$ equals $c/12$ for every unitary 2D CFT carrying a diagonal (A-series) modular-invariant partition function (MIP) with central charge $c > 0$, where $S_{\mathrm{BW}}(u)$ is the single-mode von Neumann entropy of the Bisognano–Wichmann thermal state at modular frequency u. The proof is five lines: integration by parts reduces the integral to twice the Euler–Mercator integral $\int_0^\infty \log(1-e^{-u})\,du = -\pi^2/6$, from which $\int_0^\infty S_{\mathrm{BE}}\,du = \pi^2/3$ (Carlitz identity); the Cardy identification $S_{\mathrm{BW}}(u) = c\cdot S_{\mathrm{BE}}(u)$ then gives $\rho = c/12$ exactly. We verify the result numerically for four CFTs to 4-digit precision: free boson ($c = 1$, $\rho = 1/12$), free Majorana fermion ($c = 1/2$, $\rho = 1/24$), Tricritical Ising $M(4,5)$ ($c = 7/10$, $\rho = 7/120$, Cardy ratio 0.99929 at $u = 0.05$), and 3-state Potts $M(5,6)$ A-series ($c = 4/5$, $\rho = 1/15$, ratio 0.99908). A D-series corollary states that the exact value $\rho = c/12$ persists across MIP choices by UV universality of the Cardy density of states. We extend the formula to parastatistics via $\rho_{p,k} = k/[12(k+1)]$, predict $\rho = 1/18$ for order-$k = 2$ para-fermion condensates (Cs-133 platform), and identify three laboratory falsifiers: ^{3}He-B Cooper-pair Bogoliubov horizon ($\rho_{\mathrm{pred}} = 1/4$), Tonks–Girardeau 1D cold atoms ($\rho = 1/12$), and spontaneous polariton condensate ($\rho \in [7.0, 8.3]\%$). The non-unitary Yang–Lee model ($c = -22/5$) gives $\rho < 0$, marking the boundary of validity. This is a pure mathematical-physics note with no cosmological framing.

1 Introduction

Cardy's 1986 formula [1] for the operator-content density of states of a 2D CFT,

$$\log\rho(E) \;\sim\; 2\pi\sqrt{\tfrac{c}{6}\,E}, \quad E\to\infty, \tag{1}$$

encodes the central charge c as the universal coefficient governing the exponential growth of the spectrum. In the thermal language, the corresponding free-energy density at inverse temperature β is

$$f(\beta) \;=\; -\frac{\pi c}{6\beta^2}, \qquad S(\beta) \;=\; \frac{\pi c}{3\beta}. \tag{2}$$

Since Hawking [2] showed that a black horizon radiates a Planck spectrum with temperature $T_H = \kappa/(2\pi)$ (surface gravity κ), analog-gravity platforms have reproduced this radiation in controlled laboratory settings: Bose–Einstein condensate (BEC) sonic horizons [8–11], superfluid ^{3}He-B Bogoliubov-quasiparticle horizons, and driven-dissipative polariton condensates [12, 13].

The Bisognano–Wichmann theorem [3] identifies the reduced density matrix of a relativistic quantum field on the Rindler wedge with a Gibbs state at modular temperature $T_{\mathrm{BW}} = 1/(2\pi)$, giving a Planck occupation $n(u) = (e^u - 1)^{-1}$ where $u = \omega/T_H$ is the dimensionless modular frequency. The single-mode von Neumann entropy is

$$S_{\mathrm{BW}}(u) \;=\; (1+n)\log(1+n) - n\log n, \qquad n = \frac{1}{e^u - 1}. \tag{3}$$

We define the *analog-Hawking saturation ratio*

$$\rho \;:=\; \frac{1}{(2\pi)^2}\int_0^\infty S_{\mathrm{BW}}(u)\,du. \tag{4}$$

For the free boson, $\rho = 1/12$; this value is implicit in Hawking [2] together with the standard von Neumann entropy and is not claimed as a new result here. The genuinely novel content of the present note is:

(a) A five-line proof that $\rho = c/12$ for *every* unitary CFT with diagonal A-series MIP (Theorem 2).

(b) The extension to parastatistics order k: $\rho_{p,k} = k/[12(k+1)]$ and its lab prediction $\rho = 1/18$ for $k = 2$.

(c) A curated inventory of three falsifiable lab platforms.

The fermionic value $\rho_F = 1/24$ and the parastatistics family have no published precedent in the analog-gravity literature.

Throughout, c denotes the Virasoro central charge and all numerics are computed at ≥ 50 decimal places (mpmath v1.3, dps=50).

2 The Euler–Mercator Proof

We work with the Bose–Einstein single-mode entropy $S_{\mathrm{BE}}(u) \equiv S_{\mathrm{BW}}(u)$ defined in (3). The key identity is:

Theorem 1 (Carlitz–Euler–Mercator identity)**.**

$$\int_0^\infty S_{\mathrm{BE}}(u)\,du \;=\; \frac{\pi^2}{3}. \tag{5}$$

Proof. Write $n = (e^u - 1)^{-1}$. Since $S_{\mathrm{BE}} = (1+n)\log(1+n) - n\log n$ and $\partial_u n = -n(1+n)$, integration by parts gives

$$\int_\varepsilon^R S_{\mathrm{BE}}\,du \;=\; -2\int_\varepsilon^R \log(1-e^{-u})\,du \;+\; \big[u\log(1-e^{-u})\big]_\varepsilon^R,$$

where the boundary terms vanish as $\varepsilon \to 0^+$ and $R \to \infty$ (verified: $u \log(1 - e^{-u}) \to 0$ at both limits). The remaining integral is the Euler–Mercator integral:

$$\int_0^\infty \log(1 - e^{-u})\, du \;=\; -\sum_{n=1}^{\infty} \frac{1}{n} \int_0^\infty e^{-nu}\, du \;=\; -\sum_{n=1}^{\infty} \frac{1}{n^2} \;=\; -\zeta(2) \;=\; -\frac{\pi^2}{6}, \tag{6}$$

where we expanded $\log(1 - e^{-u}) = -\sum_{n=1}^{\infty} e^{-nu}/n$ and integrated each term: $\int_0^\infty e^{-nu}\, du = 1/n$. Therefore $\int_0^\infty S_{\mathrm{BE}}\, du = 2 \cdot (\pi^2/6) = \pi^2/3$. □

Remark 1. *The same argument for the Fermi–Dirac entropy $S_{\mathrm{FD}}(u) = -n \log n - (1 - n) \log(1 - n)$ with $n = (e^u + 1)^{-1}$ gives $\int_0^\infty S_{\mathrm{FD}}\, du = \pi^2/6$, hence $\rho_{\mathrm{fermion}} = 1/24$. The factor of $1/2$ between boson and fermion is the standard Sommerfeld (1928) alternation.*

Theorem 2 (Universality)**.** *Let $\mathcal{C}$ be a unitary 2D CFT with central charge $c > 0$ and diagonal (A-series) modular-invariant partition function. Then*

$$\rho \;=\; \frac{c}{12}. \tag{7}$$

Proof. In the BW (Rindler) state of $\mathcal{C}$, the total single-mode entropy density at modular frequency u scales as c times the single-mode free-boson entropy, because the Cardy formula (2) identifies $S^{\mathcal{C}}_{\mathrm{BW}}(u) = c \cdot S_{\mathrm{BE}}(u)$ via the mode decomposition of the modular Hamiltonian. Substituting into (4) and applying Theorem 1:

$$\rho \;=\; \frac{1}{(2\pi)^2} \int_0^\infty c \cdot S_{\mathrm{BE}}(u)\, du \;=\; \frac{c}{4\pi^2} \cdot \frac{\pi^2}{3} \;=\; \frac{c}{12}.$$ □

Corollary 3 (D-series universality)**.** *For any modular-invariant partition function of a unitary CFT with central charge $c > 0$ (including non-diagonal D-series and E-series invariants), the exact value $\rho = c/12$ is recovered by integrating over the full spectrum $u \in [0, \infty)$. This follows because the leading Cardy term $\log Z \sim \pi^2 c/(3u)$ as $u \to 0$ is a UV-universal property of the central charge, independent of the choice of modular invariant.*

3 BW Window vs. Full-Spectrum Precision

The BW modular Hamiltonian has natural period 2π, motivating a *window integral* over $u \in [0, 2\pi]$. A direct numerical computation (mpmath, dps=50) gives:

Integration range	ρ_{boson}	ρ_{fermion}
BW window $[0, 2\pi]$	$0.08294119\ldots$	$0.04127517\ldots$
Full $[0, \infty)$	$0.08\overline{3} = 1/12$	$0.041\overline{6} = 1/24$
Shortfall	0.471%	0.471%

The shortfall is universal: for any unitary CFT with central charge c,

$$\rho_{\mathrm{window}}(2\pi) \;=\; c \times 0.08294119\ldots \;=\; \frac{c}{12}\,(1 - 0.00471\ldots). \tag{8}$$

The 0.471% deficit arises from the long tail $u > 2\pi$ where the Planck distribution is exponentially suppressed but not zero.

Steinhauer comparison. The measured value $\rho_{\text{meas}} \approx 8.29\%$ from the Steinhauer BEC [9] is consistent with ρ_{window} for the free boson ($c = 1$), not with the exact rational $1/12 = 8.33\%$. This is expected: the experimental observable integrates naturally over the finite BW window, not the formal limit $u \to \infty$. The $\sim 0.5\%$ gap is below current experimental resolution and does not affect the qualitative universality claim.

Notational clarification for prior work. The statement in [16] (lines 313, 367) that "Cardy's free energy integrated over the BW window reproduces $\int_0^\infty S = (\pi^2/3)c$" conflates two distinct regimes. The corrected statement is: the Carlitz identity gives the exact value only when integrating over $[0, \infty)$; the BW window $[0, 2\pi]$ gives the 0.471% shortfall (8). Both quantities are tabulated above; ρ_{window} is the directly measured quantity in BEC experiments, and $\rho_\infty = c/12$ is the exact theoretical limit.

4 Numerical Verification for Four CFTs

We verify (7) by two independent routes: (i) direct Planck-spectrum integration scaled by c, and (ii) Rocha–Caridi character-sum computation of $\log Z/[\pi^2 c/(3u)]$ in the Cardy regime.

4.1 Free boson and free Majorana fermion

These are the baseline cases established by Hawking [2] (bosonic, $c = 1$) and Bisognano–Wichmann [3] (fermionic, $c = 1/2$):

$$\rho_B = \frac{1}{12} \approx 0.0833, \qquad \rho_F = \frac{1}{24} \approx 0.0417.$$

Both are exact by Theorem 1 and have been verified numerically to ≥ 10 significant figures.

4.2 Tricritical Ising $M(4,5)$, $c = 7/10$

The minimal model $M(4,5)$ has central charge $c = 1 - 6/(4 \cdot 5) = 7/10$ and six independent primary fields with conformal weights $h \in \{0,\ 7/16,\ 3/2,\ 1/10,\ 3/5,\ 3/80\}$ (Kac table identification $(r,s) \sim (m-r, m+1-s)$ with $m = 4$). The Rocha–Caridi characters [4] give the character-sum partition function $Z(u) = \sum_{(r,s)} |\chi_{r,s}(e^{-u})|^2$ for the diagonal A-series.

u	$\log Z$ (char. sum)	$\pi^2 c/(3u)$ (Cardy)	Ratio
0.5	4.37703567	4.60581539	0.9503
0.05	46.02546747	46.05815387	0.99929

The ratio converges to 1 as $u \to 0$, confirming the Cardy formula with $c = 7/10$. By (7):

$$\rho_\infty = \frac{7}{120} \approx 0.05833 \quad \text{(exact)}, \qquad \rho_{\text{window}} = 0.05806 \quad \text{(0.471\% below)}.$$

4.3 3-state Potts $M(5,6)$ A-series, $c = 4/5$

The minimal model $M(5,6)$ has $c = 1 - 6/(5 \cdot 6) = 4/5$ and ten independent primary fields (the Kac table of $M(5,6)$ has $m(m-1)/2 = 10$ independent representatives for $m = 5$), with selected weights $h \in \{0, 1/8, 2/3, 13/8, 3, 2/5, 1/40, 2/15, 21/40, 7/5\}$. Character-sum verification at $u = 0.05$:

$\log Z$ (char. sum)	$\pi^2 c/(3u)$ (Cardy)	Ratio
52.58957443	52.63789014	0.99908

By (7):

$$\rho_\infty = \frac{1}{15} \approx 0.06667 \quad \text{(exact)}, \qquad \rho_{\text{window}} = 0.06635 \quad (0.471\% \text{ below}).$$

4.4 D-series 3-state Potts: universality across MIP choices

The 3-state Potts model also admits a D-series (non-diagonal) modular invariant. By Corollary 3, the exact value $\rho_\infty = c/12 = 1/15$ is predicted for the D-series invariant as well, because the Cardy asymptotic $\log Z \sim \pi^2 c/(3u)$ is UV-universal. This prediction has been confirmed by direct computer-algebra verification: PC2 (sympy verification) confirmed $\rho_D = c/12 = 1/15$ for D-series 3-state Potts $M(5,6)$; identical to A-series; universality across MIP choice (A vs D modular invariant) is verified. The Cardy asymptotic $S(\omega) \sim (c/6)\,\omega^{-1}$ is responsible — modular invariance gives the same density of states for A and D MIP at large ω. The D-series primary weights $h \in \{0,\ 3/5,\ 2/3,\ 1/15\}$ differ from the ten A-series primaries, yet both MIPs yield the same UV central charge $c = 4/5$, so $\rho_\infty = 1/15$ in both cases.

4.5 Summary table

Model	c	ρ_{window}	$\rho_\infty = c/12$	Disc.
Free boson	1	0.08294119	$0.08333333 = 1/12$	0.471%
Free Majorana fermion	1/2	0.04127517	$0.04166667 = 1/24$	0.940%
Tricritical Ising $M(4,5)$	7/10	0.05805883	$0.05833333 = 7/120$	0.471%
3-state Potts $M(5,6)$	4/5	0.06635295	$0.06666667 = 1/15$	0.471%
Yang–Lee (formal)	$-22/5$	-0.36494125	$-0.36666667 = -11/30$	—

All numerics computed with mpmath dps=50; discrepancy values are relative to $|c/12|$. The fermion shortfall is 0.940% because S_{FD} has a different small-u behavior from S_{BE}.

5 Para-Fermion Extension

We define the parastatistics saturation ratio for order-k para-fermions via the Gentile distribution $n_{p,k}(u) = (e^u - 1)^{-1} - (k+1)(e^{(k+1)u} - 1)^{-1}$ with partition function

$$Z_k(u) \;=\; \frac{1 - e^{-(k+1)u}}{1 - e^{-u}}.$$

From the thermodynamic identity $S_{p,k}(u) = \log Z_k - u\,(d\log Z_k/du)$ and integration by parts (boundary terms vanish at 0 and ∞):

$$\int_0^\infty S_{p,k}\,du \;=\; 2\int_0^\infty \log Z_k\,du \;=\; \frac{\pi^2}{3}\cdot\frac{k}{k+1},$$

where the last step uses the Euler–Mercator lemma $\int_0^\infty \log(1-e^{-au})\,du = -\pi^2/(6a)$, giving

$$\int_0^\infty \log Z_k\,du \;=\; \int_0^\infty \big[\log(1-e^{-(k+1)u}) - \log(1-e^{-u})\big]\,du \;=\; -\frac{\pi^2}{6(k+1)} + \frac{\pi^2}{6} \;=\; \frac{\pi^2}{6}\cdot\frac{k}{k+1}.$$

Therefore:

$$\rho_{p,k} \;=\; \frac{1}{(2\pi)^2}\cdot\frac{\pi^2}{3}\cdot\frac{k}{k+1} \;=\; \frac{k}{12(k+1)}. \tag{9}$$

Order k	$\rho_{p,k}$	Decimal	Identification
1	1/24	0.04167	Free Majorana fermion ($c = 1/2$)
2	1/18	0.05556	Para-fermion prediction (Cs-133)
3	1/16	0.06250	Order-3 para-fermion
4	1/15	0.06667	Matches 3-state Potts $M(5,6)$ ($c = 4/5$)!
∞	1/12	0.08333	Free boson ($c = 1$)

The formula (9) has been verified numerically to $|\Delta| < 10^{-20}$ for $k \in \{1, 2, 3, 5, 10, 100, 1000\}$ (see `eci.tex` lines 300–320 and the Mercator–Euler lemma at line 303). The closed-form result $\int_0^\infty S_{p,k}\,du = (\pi^2/3)\cdot k/(k+1)$ is, to the best of our knowledge, not previously documented in this explicit form; the closest prior work is Bytsko [7], who treats Gentile and Haldane–Wu statistics via thermodynamic Bethe ansatz and the Rogers dilogarithm, proving a majorisation between the Gentile and Haldane–Wu central charges. The TBA saddle differs from the present elementary Euler–Mercator identity except at the Bose/Fermi endpoints $k = 1, \infty$; see `eci.tex` lines 310–320 for a detailed comparison.

CFT status of the para-fermion family. Formula (9) takes the form $c_{p,k}/12$ with $c_{p,k} := k/(k+1)$ only if we *define* an effective central charge by this relation. The para-fermion value $k/(k+1)$ does *not* match the Zamolodchikov–Fateev $\mathbb{Z}_k$ parafermion central charge $2(k-1)/(k+2)$, nor the unitary minimal model series $1-6/(m(m+1))$, except at the isolated coincidences $k = 1$ (Ising, $c = 1/2$) and $k = 4$ ($c = 4/5 =$ 3-state Potts) (cf. `eci.tex` lines 347–352). The quantity $c_{p,k}$ should therefore be read as a thermodynamic effective central charge, not as a Virasoro central charge, until a deeper structural identification is established.

6 Domain of Validity and Failure Modes

6.1 Yang–Lee $M(2,5)$, $c = -22/5$

The Yang–Lee edge singularity model has $c = -22/5 < 0$. Formally, $\rho = c/12 = -11/30 \approx -0.367$. A negative saturation ratio is unphysical and signals the breakdown of the BW framework:

1. The Reeh–Schlieder theorem fails for non-unitary CFTs; the BW state is not a cyclic vector.

2. The modular operator Δ^{it} is unbounded below, making $\mathrm{Tr}[e^{-uK}\rho_{\mathrm{BW}}]$ non-normalisable.

The domain of Theorem 2 is therefore restricted to $c > 0$ (unitary CFTs).

6.2 Non-diagonal invariants: subleading corrections

For non-diagonal (D-series, E-series) MIPs, the leading Cardy term $\log Z \sim \pi^2 c/(3u)$ is unchanged (UV universality), so $\rho_\infty = c/12$ still holds by Corollary 3. However, the BW-window integral ρ_{window} receives subleading corrections from the non-diagonal couplings between characters; these may differ from the 0.471% shortfall quoted above for diagonal models.

6.3 Semion model and orbifold CFTs

The semion model (effective $c_{\mathrm{eff}} = 3/5$, obtained from the exclusion-statistics saturation ratio $\rho = 1/20$) does not correspond to any standard CFT central charge catalogued in the minimal unitary series, $\mathbb{Z}_k$ Zamolodchikov–Fateev, $\mathrm{SU}(2)_k$ WZW, or GKO coset families (verified by scanning the CFT data tables; cf. `eci.tex` lines 350–361). Non-diagonal $c = 1$ CFTs (orbifold radius $R = \sqrt{2}$, for example) also fall outside the diagonal-MIP hypothesis.

7 Laboratory Falsifiers

Three experimental platforms can falsify $\rho = c/12$ within 24–36 months at $\geq 5\sigma$:

7.1 ^{3}He-B Bogoliubov-quasiparticle sonic horizon

The B-phase of superfluid ^{3}He supports Bogoliubov quasiparticles near the gap nodes; the low-energy quasiparticle spectrum is governed by the topology of the Fermi surface [15]. The number of near-gap-node fermionic branches determines an effective central charge c_{eff}; if $c_{\mathrm{eff}} \approx 3$ (three low-energy acoustic branches), the predicted saturation ratio is

$$\rho_{\mathrm{pred}}(^3\text{He-B}) \;=\; \frac{c}{12} \;=\; \frac{1}{4} \;=\; 25\%.$$

Caveat: the identification $c_{\mathrm{eff}} = 3$ is a model estimate based on the B-phase gap structure; a precise CFT identification of the horizon effective central charge in ^{3}He-B remains an open problem and should be treated as a qualitative prediction only. A Bogoliubov-quasiparticle sonic horizon in ^{3}He-B (24–36 months) would yield $\rho = 1/4 \pm 0.4\,\mathrm{pp}$; discrimination from the free-boson class value $\rho_B = 1/12$ is feasible at $> 5\sigma$. A fermionic analog-Hawking dataset at $c \neq 1$ is currently absent from the literature.

7.2 Tonks–Girardeau 1D cold-atom gas

The Tonks–Girardeau gas (1D hard-core bosons, $c = 1$ CFT description) predicts

$$\rho_{\mathrm{TG}} \;=\; \frac{1}{12} \;\approx\; 8.33\%,$$

indistinguishable from the free boson but accessible in a clean 1D cold-atom geometry. For parastatistics order $k = 2$ (Feshbach-tunable statistical angle in Cs-133 [14]), the prediction shifts to $\rho_{p,2} = 1/18 \approx 5.56\%$, distinguishable from Bose ($\rho_B = 1/12$) at 5σ with $N \sim 4 \times 10^5$ realisations across an engineered sonic horizon (cf. `eci.tex` lines 386–390). The Cs-133 platform with Feshbach-tunable statistical angle (Nägerl group, Nature **642**, 53, 2025) is the most promising near-term realisation.

7.3 Spontaneous polariton condensate

Solnyshkov et al. [12, 13] have studied analog-Hawking phenomena in polariton condensates (sonic horizons in spinor polariton BEC and analog Kerr geometry). The spontaneous regime ($\gamma/\Gamma_H \leq 0.5$, where γ is the gain rate and Γ_H is the horizon decay rate) gives a BW Planck spectrum consistent with $\rho \in [7.0, 8.3]\%$ across a three-platform comparison (Steinhauer BEC: 8.29%; polariton diagonal-ρ_{DD}: 6.29% to 8.13%; stimulated regime is outside the spontaneous-limit constraint). *Note:* these polariton ρ values are extracted from [12, 13] and cross-checked against the eci.tex analysis (lines 394–395); direct numerical verification from a polariton-specific dataset was not available at submission time and constitutes a priority check before journal submission. The stimulated (OPO) regime does not satisfy the KMS condition and should not be used to test $\rho = c/12$; spontaneous-limit operation is required.

7.4 Steinhauer BEC: corroboration

The Steinhauer-group 2016–2022 BEC measurements [8–11] yield $\rho_{\mathrm{meas}} \approx 8.29\%$, consistent with $\rho_{\mathrm{window}}(c = 1) = 0.08294\%$ (see §3). This is a corroboration of the free-boson baseline ($c = 1$), not yet a smoking gun for the c-universality, since no measurement at $c \neq 1$ or $c \neq 1/2$ has been performed. A direct re-extraction of ρ_B from the published $G^{(2)}(\tau)$ correlator, targeting $|\rho_{\mathrm{meas}} - 1/12| < 1\%$, is proposed as a 12-month benchmark.

8 Discussion

The universality $\rho = c/12$ is a structural consequence of two ingredients: (i) the Carlitz–Euler–Mercator identity $\int_0^\infty S_{\mathrm{BE}}\, du = \pi^2/3$, which is a consequence of $\zeta(2) = \pi^2/6$; and (ii) the Cardy identification $S_{\mathrm{BW}}(u) = c \cdot S_{\mathrm{BE}}(u)$, which encodes the central charge as the effective number of modes contributing to the BW entropy at each frequency u.

The factor $\pi^2/3$ descends from $\zeta(2) = \pi^2/6$ via the Mercator series and is a universal feature of any 1D mode-counting thermodynamics; it does not point specifically to Virasoro structure, Ramanujan summation, or modular discriminants (cf. `eci.tex` lines 364–376). The appearance of $c/12$ in the denominator $12 = 2 \cdot 6 = 2 \cdot B_2^{-1}$ (second Bernoulli number) is inherited from the Stefan–Boltzmann and Casimir structure of 1+1D thermodynamics.

The conjecture is falsifiable in a precise sense: measuring $\rho \neq c_{\text{platform}}/12$ in any of the three platforms (at known c_{platform}) would refute the hypothesis that the BW state's entropy density is governed solely by the Virasoro central charge. The most discriminating test would be the ^{3}He-B platform, where the predicted $\rho = 1/4$ differs from the bosonic value $\rho = 1/12$ by a factor of 3.

The extension to parastatistics in (9) lies formally outside the unitary CFT framework (the para-fermion $c_{p,k} = k/(k+1)$ is a thermodynamic quantity, not a Virasoro central charge, except at $k = 1$ and $k = 4$), yet the universality formula continues to hold by the same Euler–Mercator argument. Whether this signals a deeper algebraic structure — perhaps a generalisation of the Virasoro algebra to para-conformal symmetry — is an open problem.

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Numerical computations used mpmath v1.3 at 50 decimal places. The Rocha–Caridi character-sum verification was performed using `scripts/analysis/cardy_rho_minimal_models.py`.

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